

Benchmark Verdicts for Analog Ising Machines Invert Under Tuning: A Negative Result and a Reproducible Protocol

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Abstract

We report a systematic simulation study of a measurement-feedback parametron Ising machine (an “analog” or coherent-Ising-machine-class solver) benchmarked against discrete Simulated Bifurcation (dSBM). Our central finding is methodological and negative: on identical instances at matched compute, the competitive verdict *inverted twice* depending solely on which solver received parameter tuning. Untuned analog versus untuned digital gave dSBM a 5.9–7.4× advantage; tuned analog versus untuned digital gave the analog machine a 2.3–6.8× advantage; tuning both — the only defensible comparison — gave near-parity, and we further find that the *sign* of the residual difference depends on how large a tuning grid each side receives. We argue that untuned baselines are a systemic hazard in this literature and propose a minimal protocol to avoid them. We additionally report: (i) a measured time-to-solution scaling of $TTS \sim N^{2.115}$ (95% CI [1.93, 2.30], $\Delta AIC = 9.97$ favouring a power law over exponential growth); (ii) four algorithmic modifications yielding a compounded $\approx 24\times$ improvement over a schedule frozen at small N , of which the largest arises from a parameter never previously varied; and (iii) an analysis showing that the architecture’s operation-count advantage is negligible in measurement feedback form, because the coupling multiply is digital in both machines. All simulation code, the hashed benchmark instance library, and the full result set are released.

1 Introduction

Analog and physics-inspired Ising machines are widely proposed as accelerators for combinatorial optimization. Coherent Ising machines with measurement feedback have been demonstrated at large spin counts at room temperature, and a substantial literature reports favourable comparisons against digital solvers.

A recurring weakness in such comparisons is that the proposed machine is typically tuned — often extensively — while the digital baseline is run at default or literature settings. We encountered this failure mode directly and severely: over the course of this study our own competitive conclusion reversed twice, purely as a function of which side had been tuned, before settling at near-parity. We report this as the primary result, because we believe it is common and largely unreported.

We study a parametron (degenerate parametric oscillator) network with digital measurement feedback, the architecture of Goto’s parametron [1] combined with the measurement-feedback scheme of modern coherent Ising machines [2, 3] and Leleu-type amplitude error correction [4]. Our digital comparator is discrete Simulated Bifurcation [5, 6].

Our contributions:

1. A demonstration that benchmark verdicts in this class invert under tuning, with the full sequence documented (Sec. 3).
2. A properly controlled near-parity result (Sec. 4).
3. A measured scaling law with confidence intervals and model comparison (Sec. 5).
4. Four algorithmic findings, including one apparently novel method (Sec. 6).
5. An operation-count analysis identifying where, if anywhere, an advantage can exist for this architecture (Sec. 7).

We emphasise at the outset that all results here are simulation. No hardware was built. Claims about physical implementations are explicitly labelled as projections in Sec. 8.

2 Methods

2.1 Model

Each spin i is represented by a slow amplitude envelope x_i obeying a rotating-frame stochastic differential equation, integrated by Euler–Maruyama:

$$dx_i = (p(t) - 1 - x_i^2)x_i + \epsilon f_i dt + \sigma dW_i, \quad (1)$$

$$f_i = e_i X_j$$

$$\tilde{J}_{ij} x_j(t - \tau_{\text{lag}}), \quad (2)$$

$$de_i = -\beta e_i(x_i^2 - a(t)) dt, \quad a(t) = \max(p(t) - 1, 0.05), \quad (3)$$

where $p(t)$ is the pump ramp, $\tilde{J}$ is row-sum normalised, e_i are Leleu-style amplitude error-correction variables, and the feedback uses a measurement delayed by one update interval. Spins are read out as $s_i = \text{sign}(x_i)$.

2.2 Comparator

Discrete Simulated Bifurcation [6] with inelastic walls ($|x| \leq 1$, momentum reset on contact) and $\text{sign}(x)$ coupling, using a standard coupling normalisation.

2.3 Cost metric

We count *matrix–vector products* (equivalently, multiply–accumulate operations), not wall-clock time or integration steps. This is the fair unit because it is the dominant work in both algorithms and, critically, is performed digitally in both: the analog machine’s coupling field is computed on an FPGA and dSBM’s on a GPU. Time-to-solution is $\text{TTS}_{99} = C \ln(0.01)/\ln(1 - p)$, with C the cost of one anneal and p the per-anneal success probability.

2.4 Instances and reference

Dense random ± 1 Ising instances, $J_{ij} \in \{-1, +1\}$, $J_{ii} = 0$. For $N \leq 22$ ground states were obtained by exhaustive enumeration; above that the reference is the best energy found across all runs of that instance. We verified this reference is not loose: on 80 instances at $N = 12, 16, 20, 22$, the best-found energy equalled the exact ground state in every case (inflation factor 1.00).

2.5 Statistics

Unless stated otherwise, 25 instances per problem size, medians across instances, and bootstrap confidence intervals resampled over instances. Success probabilities are computed over 100–300 independent anneals per instance.

3 Benchmark verdicts invert under tuning

Table 1 shows the same comparison, on the same instances, at matched matrix-multiply budget, evaluated three times during this study.

Comparison $N = 32$ $N = 64$ Status

Untuned analog vs. untuned dSBM dSBM 5.9× dSBM 7.4× analog untuned
 Tuned analog vs. untuned dSBM analog 6.8× analog 2.3× dSBM untuned Both
 tuned near-parity (see Sec. 4) defensible

Table 1: The competitive verdict inverts twice depending only on which solver is tuned. Only the third row is a defensible comparison. Both solvers turned out to be running substantially longer than optimal in the first two rows.

The magnitude matters: the first two rows differ by a factor of roughly 40 at $N = 32$ and would support opposite conclusions in a paper abstract. Neither was produced dishonestly; each was the result of tuning one side and not the other, which is the default outcome when a research group benchmarks its own system against a literature baseline.

Proposed minimal protocol. We adopt, and suggest, the following rule: *no comparison is reportable unless the same tuning budget has been applied to every algorithm in it, and the settings used for each are stated alongside the result, together with the size of the parameter grid searched for each.* An untuned baseline is not a baseline, and a tuned baseline whose search budget is unstated is only marginally better (Sec. 4). We further recommend counting cost in operations rather than wall-clock time, since wall-clock comparisons across substrates conflate algorithmic quality with implementation effort.

4 Near-parity result

With run length optimised for both solvers, the two methods are at near-parity. We deliberately do not report a single ratio here, because the residual difference proved smaller than the effect of the tuning grid itself: two independent tuning efforts on the same instance family gave dSBM ahead by 1.13× at $N = 32$ in one case, and the analog machine ahead by 1.72× in the other, with an exact tie at $N = 64$. Both efforts searched comparable but not identical grids.

We treat this as a second-order instance of the paper’s main finding rather than a contradiction of it: if the sign of a competitive verdict can be changed by the size of a tuning grid, a single reported ratio from a single tuning effort carries little information. The defensible claim is near-parity, and any stronger claim in either direction requires the tuning budget searched — not only the final settings — to be reported alongside the result. Fitted scaling exponents under equal tuning are $N^{2.72}$ (analog) versus $N^{2.32}$ (dSBM), so the deficit widens slowly with problem

size.

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We regard near-parity against a mature, corporate-refined production algorithm as a substantive result for an architecture that does not physically exist. We also regard it as insufficient to support an energy claim, for reasons given in Sec. 7.

5 Scaling

Time-to-solution was measured at six sizes ($N = 8-64$), 25 instances each, with anneal length optimised independently at every size (Table 2).

N	Optimal anneal p	at optimum TTS (matmuls)	Local slope
8	3,000	0.97	394 —
16	3,000	0.51	1,937 2.30
24	10,000	0.75	3,322 1.33
32	10,000	0.52	6,274 2.21
48	10,000	0.27	14,633 2.09
64	10,000	0.11	39,518 3.45

Table 2: Scaling scan, 25 instances per size.

A power law fits substantially better than an exponential: $TTS \sim N^{2.115}$ (bootstrap 95% CI [1.93, 2.30], AIC = -18.21) versus $TTS \sim \exp(0.0748N)$ (AIC = -8.24). With $\Delta AIC = 9.97$ the exponential model has relative likelihood ≈ 0.007 . Growth over the measured range is therefore polynomial.

Two caveats. First, the local slope in the final interval ($48 \rightarrow 64$) is 3.45, the steepest in the scan; this is consistent either with noise at 25 instances or with genuine curvature, and data ending at $N = 64$ cannot distinguish them. Second, exponent estimates converged only as instance count rose ($2.48 \rightarrow 2.11 \rightarrow 2.01 \rightarrow 2.115$ at 3, 5, 10, 25 instances), with the confidence interval halving between 10 and 25. We recommend ≥ 25 instances per size for scaling claims in this class.

6 Algorithmic findings

Four parameters that had been fixed by inheritance rather than measurement yielded substantial gains (Table 3).

Modification Gain Note

Update interval $10 \rightarrow 100 \approx 8\times$ Physics between updates is free Inelastic wall at $0.4A \approx 2\times$ Adopted from dSBM; one limiter per cell Ramp exponent $\gamma = 1 \rightarrow 5 \approx 2\times$ Zero cost; waveform change only Confidence-gated feedback $1.45-1.94\times$ Novel; $N \leq 64$ only

Table 3: Algorithmic modifications, each measured with the comparator held fixed.

Confidence-gated feedback. Once a spin’s amplitude is pinned at the limiter and its sign has been stable across consecutive updates, its contribution to every other spin’s field is constant. We compute it once, add it to a cached vector, and remove the spin from the active set; the matrix multiply then runs over active spins only and shrinks as the anneal proceeds. This exploits a

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property specific to analog machines — amplitude carries a confidence signal that a purely binary solver does not possess. Measured MAC reduction is 2.0–2.2× with a net TTS improvement of 1.45–1.94× at $N \leq 64$. The method fails at $N = 128$, where baseline success probability is already 0.01 and premature freezing destroys the remaining chance.

Two further variants were tested and closed: parallel tempering across the batch axis (0.64–0.68×, worse) and cyclic pumping ($\leq 0.13\times$, much worse). Predictive extrapolation of the feedback was strongly harmful ($p : 0.805 \rightarrow 0.105$), as it amplifies measurement noise.

7 Where an advantage can and cannot exist

In measurement-feedback form the coupling product Jx is computed digitally, on an FPGA, exactly as dSBM computes it on a GPU. There is therefore no per-operation energy advantage available to offset an operation-count deficit, and Sec. 4 shows there is no operation-count surplus either. We conclude that the measurement-feedback architecture has no defensible energy claim on dense problems, and we note that energy comparisons in this literature frequently assume operation parity without testing it.

The remaining possibility is architectural. If the coupling is physically wired rather than computed — possible when J is fixed and sparse — the anneal requires no digital work at all beyond readout. Under that assumption we compute a 150–180× reduction in digital operations on lattice instances. Separately, we verified in simulation that such an array anneals successfully *without* measurement feedback or amplitude error correction, which the wired configuration cannot support: autonomous operation matched or exceeded measurement feedback on lattice instances ($p = 0.292$ vs. 0.254 at 64 spins).

We stress that this is an architectural analysis, not a measurement of hardware.

8 Limitations

- ^ All results are simulation. No hardware was constructed. Every statement about physical implementations, settling times, or energy is a projection from the model.
- ^ Arm A versus Arm B is untested. The central hypothesis of this architecture class — that physical analog dynamics outperforms a digital simulation of the same dynamics — cannot be evaluated without hardware and is not evaluated here.
- ^ Instance class. All dense results use random ± 1 instances; structured, sparse, and application derived instances may rank algorithms differently.
- ^ Comparator implementation. Our dSBM is a reference implementation, not the authors’ optimised version, and may understate its performance.
- ^ Sizes. Scaling data ends at $N = 64$; extrapolation beyond that is not evidence.

9 Conclusion

The competitive standing of an analog Ising machine against a tuned digital solver, on dense random instances, is near-parity — and we could have reported either a large win or a large loss from the same data by tuning one side. We believe this is the most useful thing we can contribute: the field should treat untuned baselines as a defect, state tuning budgets explicitly, and count cost in

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operations. We release all code, instances and results so that our numbers can be checked and our errors found.

Data and code availability

All simulation code, the benchmark instance library (100 instances, $N = 8$, with brute-forced ground states, SHA-256 b8c6050b2edf1ed40d02ae97e0e37b39ea60661822ccc5afa3c40cfd3c48812), and the complete result set are available at <https://github.com/shmuelbiller1/apim-benchmark-study.git>.

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